Utica University | Department of Computer Science | Fall 2026

CSC 618: Information Retrieval | Week 10 Assignment

Submission 3: Part C, Questions on the Real-Dataset Experiment

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I compared TF-IDF, BM25 and a Dirichlet Language Model in Apache Lucene on the Cranfield collection (1400 documents, 225 queries). Full methods and code are in Submission 4.

Table 1: Key results (mean over 225 queries, default parameters).

Model	MAP	MRR	P@3	P@10	nDCG@10	Recall (full list)
TF-IDF	0.2931	0.5327	0.3452	0.2244	0.3690	0.9513
BM25	0.2952	0.5204	0.3570	0.2284	0.3741	0.9523
LM-Dirichlet (μ=2000)	0.2341	0.4557	0.2726	0.1840	0.3041	0.9515

1. Did the controlled-dataset code work on the real dataset? What changed?

Yes. The metric functions in `ir_metrics.py` ran without modification, and an independent library (ranx) gave identical values to six decimals. The changes were all around those functions:

Change	Why
Loaders for qrels and runs	Part B data was typed into Python; Part C needs TREC-format files
Renumbering query IDs	Query numbers in the file are 1, 2, 4, 8, … up to 365, but qrels use 1–225 by position. Without the fix, BM25 MAP falls from 0.2952 to 0.0090, with no error raised. This was the most important change
Added cut-offs	Precision, Recall and F1 on a list of up to 1000 are nearly meaningless (Precision 0.0081), so I added them at rank 10
nDCG for all queries with binary gain	Part B had grades for Q1 only; the copy of the qrels I could obtain is binary

2. Did the real-world results match my expectations?

Partly. I expected BM25 to clearly beat TF-IDF and the Language Model to be close to BM25.

- **BM25 vs TF-IDF: smaller than expected.** BM25 has the higher MAP by only 0.0021, which is not significant (randomization $p = 0.7492$). TF-IDF has a slightly higher MRR and is the outright best model by AP on more queries (99 vs 75). Lucene's TF-IDF includes length normalisation, which probably removes much of the difference that a basic TF-IDF would show.
- **LM-Dirichlet: worse than expected.** Its MAP is 0.0611 below BM25 ($p < 10^{-5}$). The default μ=2000 is far above the mean document length of 102 terms. Lower μ helped (MAP 0.2757 at μ=50) but did not reach BM25. Lucene's implementation also scores only documents containing a query term and clamps negative scores to zero, so it is not a pure query-likelihood model. I did not test separately how much that contributes.
- **Absolute scores: as expected for a hard small collection.** A MAP near 0.30 and P@10 near 0.23 were reasonable: queries average 7.2 relevant documents, so P@10 cannot be high for many queries.

3. Which metrics were most informative, and why?

Metric	Value in this experiment
MAP	**Most informative.** Uses every relevant document's position, and clearly separated the Language Model
nDCG@10	Same conclusion as MAP, focused on the first page. Less informative than usual, as gains were binary
MRR	Added a distinct view: TF-IDF and BM25 swap order on it, though not significantly
P@10, Recall@10	Easy to interpret, but limited by the number of relevant documents per query (range 1–39)
Full-list P, R, F1	**Least informative.** Nearly identical across models (Recall 0.951–0.952), because these lists include almost all documents matching any query term

The per-query paired tests were as useful as the metrics themselves: without them, the BM25 and TF-IDF means could have been taken as a real difference.

4. What are the main limitations of my evaluation?

1. **Old, narrow collection.** 1960s aeronautics abstracts with verbose questions (about 18 words) do not represent modern web or enterprise search.
2. **Binary relevance.** The original Cranfield grades were not in the copy I could access, so nDCG did not use graded relevance here. Rows labelled "no interest" in the source were treated as non-relevant.
3. **Incomplete judgments.** Unjudged documents count as non-relevant, which can understate any model.
4. **Parameter choice.** The main comparison uses default parameters. The sensitivity results were chosen after viewing test queries, with no separate tuning set, so they are not clean performance estimates.
5. **One pipeline.** Results depend on Lucene's implementations and on `EnglishAnalyzer`, whose small stop list keeps words such as `what`.
6. **No user data.** These offline metrics say nothing about speed, satisfaction or real click behaviour.